

A different, strikingly elegant, “reverse search” enumeration method was very recently described by Avis & Fukuda [31]. If we choose any generic linear function, then the simplex algorithm (cf. Section 3.2) with “Bland’s rule” finds a path from every vertex of the polytope to the maximal vertex of the polytope. Now if we have a simple polytope, these paths form a tree that connects the vertices, and which can be searched easily. This yields a very effective algorithm — try Avis [27, 28]. In the nonsimple case, one has to search a tree on the (huge) set of all feasible bases; with some extra care one can detect those bases that are lexicographically first at a vertex. (See also Rote [466].) Avis [27] has reported the successful solution of very large convex hull problems with reverse search; see also [144].

Avis, Bremner & Seidel [29, 30, 131], however, construct and analyze classes of “bad” test examples for different types of convex hull algorithms. In particular, products of cyclic polytopes of the form $C_d(n)^d$ seem to be “universally” bad for *all* known types of convex hull algorithms.

Problems and Exercises

- 1.0 If we try to restrict the proof of Theorem 1.2 to polytopes, where does this fail? In other words, where do we use a construction that necessarily takes us from polytopes to unbounded polyhedra?
- 1.1 Do experiments with the methods of this chapter on some examples:
 - (i) Compute the vertices of the 2-dimensional example of Section 1.2. Check carefully that everything you get actually is a vertex.
 - (ii) Compute defining inequalities for the cyclic polytope $C_3(6)$, for example for $t_i = i$. For every inequality, find the set of vertices that satisfy it with equality.
 - (iii) Find the facets of the 4-polytopes of the Exercises 4.8.6 and 4.8.15 from Grünbaum [252, pp. 64,65].
 - (iv) These were Mickey-Mouse** examples (i.e., very small).

For more realistic ones, find all the facets of the traveling salesman polytopes $Q_T(6)$, $Q_T(7)$, $Q_T(8)$, and of the asymmetric travelling salesman polytopes $Q'_T(5)$ and $Q'_T(6)$.

(Better use a computer; $Q_T(8)$ is a 20-dimensional polytope with 2520 vertices and 194,187 facets [153]. The polytope $Q_T(9)$ has 42,104,442 facets; for $Q_T(10)$ Christof found 51,043,900,866 facets and conjectures that they give a complete description — see [152] and [154]. Similarly, Euler & Le Verge [200] have derived a description of $Q'_T(6)$: a 19-dimensional polytope with 120 vertices and 319,015 facets.)

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- 1.2 Describe a Fourier-Motzkin elimination method to solve strict inequality systems $\{\mathbf{x} : A\mathbf{x} < \mathbf{z}\}$. Use it to prove a representation theorem for “open polyhedra.”
- 1.3 Show that if $C = \text{cone}(W)$ is any cone in $\mathbb{R}^{d+1}$ generated by arbitrary vectors $\mathbf{w}_i$ (not necessarily with $w_{i0} \geq 0$), then $\{\mathbf{x} \in \mathbb{R}^d : \begin{pmatrix} 1 \\ \mathbf{x} \end{pmatrix} \in C\}$ is a $\mathcal{V}$ -polyhedron.
- 1.4 State and prove a Farkas lemma for systems of the form $A\mathbf{x} \leq \mathbf{z}$, $\mathbf{x} \geq \mathbf{0}$.
- 1.5 Prove the following Farkas lemma for equality and inequality constraints: for compatible matrices A, B, C and vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ **either** there exists a solution vector $\mathbf{x}$ for

$$A\mathbf{x} = \mathbf{u}, \quad B\mathbf{x} \geq \mathbf{v}, \quad C\mathbf{x} \leq \mathbf{w},$$

or there exist row vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ with

$$\mathbf{a}A + \mathbf{b}B + \mathbf{c}C = \mathbf{0}, \quad \mathbf{b} \leq \mathbf{0}, \quad \mathbf{c} \geq \mathbf{0}, \quad \mathbf{a}\mathbf{u} + \mathbf{b}\mathbf{v} + \mathbf{c}\mathbf{w} < 0.$$

- 1.6 Prove the following general Farkas lemma for equality and inequality constraints: for compatible matrices A, B, C, D and vectors $\mathbf{z}, \mathbf{w}$ **either** there exist solution vectors $\mathbf{x}, \mathbf{y}$ for

$$A\mathbf{x} + B\mathbf{y} \leq \mathbf{z}, \quad C\mathbf{x} + D\mathbf{y} = \mathbf{w}, \quad \mathbf{x} \geq \mathbf{0},$$

or there exist row vectors $\mathbf{c}, \mathbf{d}$ with

$$\mathbf{c}A + \mathbf{d}C \geq \mathbf{0}, \quad \mathbf{c}B + \mathbf{d}D = \mathbf{0}, \quad \mathbf{c} \geq \mathbf{0}, \quad \mathbf{c}\mathbf{z} + \mathbf{d}\mathbf{w} < 0.$$

- 1.7 State and prove a version of Carathéodory’s theorem for convex-conical combinations.
- 1.8 *Transportation polytopes* have the form

$$P(d : \mathbf{a}, \mathbf{b}) = \left\{ X \in \mathbb{R}^{d \times d} : \begin{array}{ll} x_{ij} \geq 0 & \text{for } 1 \leq i, j \leq d, \\ \sum_{k=1}^d x_{ik} = a_i & \text{for } 1 \leq i \leq d, \\ \sum_{k=1}^d x_{kj} = b_j & \text{for } 1 \leq j \leq d. \end{array} \right\}$$

Study transportation polytopes. Determine the dimension. Interpret vertices and facets. Show that the Birkhoff polytopes are special transportation polytopes. Show that non-empty transportation polytopes $P(d : \mathbf{a}, \mathbf{b})$ have canonical center points, given by $x_{ij} = \frac{a_i b_j}{\Lambda}$ for all i and j , where $\Lambda := \sum_{k=1}^d a_k = \sum_{k=1}^d b_k$.

- 1.9 Interpret the Farkas lemma IV as a statement about polyhedra, and observe that it follows “trivially” from the equivalence of $\mathcal{V}$ - and $\mathcal{H}$ -polyhedra (Theorem 1.2). Derive Farkas lemma II from Farkas lemma IV, and then Farkas lemma I from Farkas lemma II.
(This alternative route through the jungle of Farkas lemmas was suggested by Joe Bonin.)

2

Faces of Polytopes

In this lecture we will discuss faces, and the face lattice. Here we restrict our attention entirely to polytopes, although nearly everything can quite easily be generalized to polyhedra; see the exercises.

I hope that the reader enjoys the ease with which we will get the results in this lecture. In fact, nearly all results are “geometrically clear,” and as far as we need algebra to verify them, we get by with straightforward computations and the Farkas lemmas.

2.1 Vertices, Faces, and Facets

Definition 2.1. Let $P \subseteq \mathbb{R}^d$ be a convex polytope. A linear inequality $\mathbf{c}\mathbf{x} \leq c_0$ is *valid* for P if it is satisfied for all points $\mathbf{x} \in P$. A *face* of P is any set of the form

$$F = P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}$$

where $\mathbf{c}\mathbf{x} \leq c_0$ is a valid inequality for P . The *dimension* of a face is the dimension of its affine hull: $\dim(F) := \dim(\text{aff}(F))$.

For the valid inequality $\mathbf{0}\mathbf{x} \leq 0$, we get that P itself is a face of P . All other faces of P , satisfying $F \subset P$, are called *proper* faces.

For the inequality $\mathbf{0}\mathbf{x} \leq 1$, we see that $\emptyset$ is always a face of P .

The faces of dimensions 0, 1, $\dim(P) - 2$, and $\dim(P) - 1$ are called *vertices*, *edges*, *ridges*, and *facets*, respectively. Thus, in particular, the vertices are the minimal nonempty faces, and the facets are the maximal proper faces. The set of all vertices of P , the *vertex set*, will be denoted by $\text{vert}(P)$.